

Amrutesh Arun

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WORK EXPERIENCE

RENCI, University of North Carolina – Chapel Hill

May 2025 – Present

Web Application Developer Intern

Chapel Hill, NC (USA)

- Developed a Node.js/Express/TypeScript microservice backend with PostgreSQL (Prisma) for ROBOKOP web application, implementing GitHub, Google, and passkey-based authentication to enable users to save and load queries.
- Integrated authentication into a React frontend with protected routes, profile management, and query UI, while improving developer workflow through hot-reload support and bug fixes.
- Upgraded the entire codebase from React v16 to v19 and changed the codebase from JavaScript to TypeScript.
- Containerized the backend using Docker for consistent development and production environments.

University of North Carolina at Charlotte

August 2024 – August 2025

Graduate Research Assistant | October 2024 - Present

Charlotte, NC (USA)

- Assisted in researching and developing a Large Language Model based HIV awareness mobile application in collaboration with UNC School of Nursing, Yale University and Duke University.

Graduate Teaching Assistant | August 2024 - Present

Charlotte, NC (USA)

- Assisted in delivering course content, focusing on full-stack development using NodeJS, EJS and MongoDB.
- Developed an entire React course from scratch, which included creating videos on React, assignments, and projects.

Wasserstoff

August 2022 – November 2023

Software Developer

Gurugram, Haryana (IN)

- Responsible for development of 10+ cross-platform application in blockchain and e-commerce sectors, reaching a user base of 10,000+, collaborated with UI/UX designers to create SEO-optimized pages, boosting organic traffic by 20%.
- Created responsive web and mobile apps using React, NextJS, React Native and Expo frameworks Engineered scalable APIs and real-time features using NodeJS, ExpressJS, and Socket.IO, facilitating seamless trading and currencies.

upGrad Jeet

August 2021 – July 2022

Software Developer

Bangalore, Karnataka (IN)

- Developed & maintained admin panel, revamped the test engine utilized by over 5,000 educators and learners.
- Integrated error tracking and performance monitoring with Sentry, reducing user-reported issues by 40%.
- Implemented user engagement tracking via MoEngage & Google Analytics, driving improvements based on data insights.

EDUCATION

University of North Carolina- Charlotte

January 2024 – Present

Master's of Science in Information Technology

Charlotte, NC (USA)

Visvesvaraya Technological University

August 2017 – July 2021

Bachelor of Engineering in Information Science

Bangalore, Karnataka (IN)

PROJECTS

WorkQuest ([Project Link](#))

- A comprehensive redesign of an existing job search platform inspired by Craigslist Jobs, aimed at improving user experience, accessibility, and functionality (**Figma, JavaScript, React, Tailwind, Node, MongoDB**).

NYC Airport Traffic Analysis ([Project Link](#))

- Analyzed 20+ years of passenger & freight to identify seasonal trends (**Python, Pandas, NumPy, Matplotlib, Plotly**)

PayShare ([Project Link](#))

- A platform designed to simplify expense sharing among friends and groups. It helps users to split bills, track expenses, and settle debts effortlessly. (**TypeScript, React Native, Redux Toolkit, Node, Express, MongoDB**)

CERTIFICATIONS, SKILLS & INTERESTS

- **Skills:** JavaScript, TypeScript, HTML, CSS, Java, Python, C, NoSQL, SQL, React, Next.js, Node.js, Flask, Django, React Native, Expo, Redux, Tanstack, Tailwind, Material UI, Framer, Jest, AWS, GraphQL, Figma, Postman, GitHub, VS Code
- **Certifications:** [AWS Certified Solutions Architect – Associate](#), [Google IT Automation with Python](#), [Foundation of UX](#)
- **Achievements:** First place in multiple global hackathons including Courier hacks: Feature Focus and Python Programs
- **Paper Publications:** “Design of Hierarchical Data Aggregation for Internet of Underwater Things” at ICGCCP-2021.